

When Fairness meets Bias: a Debiased Framework for Fairness aware Top-N Recommendation

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ABSTRACT

Fairness in the recommendation domain has recently attracted increasing attention due to more and more concerns about the algorithm discrimination and ethics. While recent years have witnessed many promising fairness aware recommender models, an important problem has been largely ignored, that is, the fairness can be biased due to the user personalized selection tendencies or the non-uniform item exposure probabilities. To study this problem, in this paper, we formally define a novel task named as unbiased fairness aware Top-N recommendation. For solving this task, we firstly define an ideal loss function based on all the user-item pairs. Considering that, in real-world datasets, only a small number of user-item interactions can be observed, we then approximate the above ideal loss with a more tractable objective based on the inverse propensity score (IPS). Since the recommendation datasets can be noisy and quite sparse, which brings difficulties for accurately estimating the IPS, we propose to optimize the objective in an IPS range instead of a specific point, which improves the model fault tolerance capability. In order to make our model more applicable to the commonly studied Top-N recommendation, we soften the ranking metrics such as Precision, Hit-Ratio, and NDCG to derive a fully differentiable framework. We conduct extensive experiments to demonstrate the effectiveness of our model based on four real-world datasets.

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CCS CONCEPTS

• Information systems → Recommender systems.

KEYWORDS

recommendation system, fairness aware recommendation

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1 INTRODUCTION

Recommender system, as an effective remedy for information overloading, has greatly facilitated people's daily life and production. In the past few decades, improving accuracy has long been the major objective for the recommender models. However, recommender system basically aims to serve human beings, thus ethical concerns cannot be ignored, among which fairness is an important aspect [27] and has recently attracted increasing attention. To build fairness aware recommenders, people have proposed a lot of strategies, for example, [37] defines four types of fairness constraints to regularize the matrix factorization model. [31, 32] achieve fairness by removing the sensitive information based on adversarial training. [19] considers the personalization of the users and defines counterfactual fairness based on causal notions. [12] pulls the performances between the advantaged and disadvantaged groups through distribution matching.

While these models have achieved many successes, an important problem has been largely ignored, that is, the fairness can be biased due to the non-uniform observational datasets. As illustrated in Figure 1, suppose we have two user groups A and B. *Ideally, a fair model should provide equal recommendation performance for the users in A and B on all the items.* However, in real-world scenarios, the observed datasets are usually biased. For example, suppose the users in the system all like horror movies better than comedy ones. Then, in their interaction records, we can observe a much

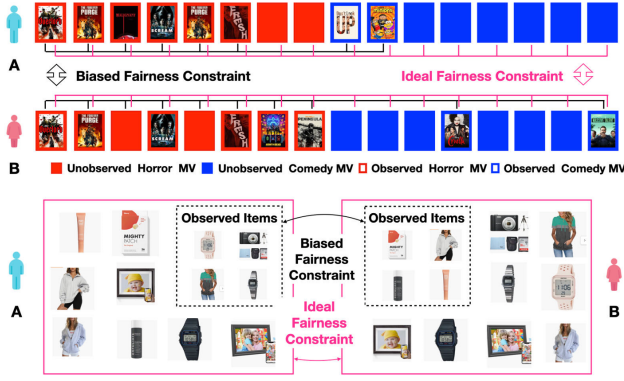


Figure 1: Illustration of the differences between the biased and ideal fairness constraints (best view in color).

higher percentage of horror items. In this case, if we directly impose fairness constraints on the observed data, the performance gap on the (underrepresented) comedy movies may be not well controlled, and if we evaluate the performance on the comedy movies, the results for group A and B can be not fair. Similar phenomena may also happen in the bottom case, where the interactions of group A are more about digital products, while the users in group B are more interested in beauty items. By a traditional fairness aware model, we may lower the performance gap between group A and B on the digital and beauty items, respectively. However, for the clothes, which are seldomly purchased by group A and B, there is no explicit constraint to guarantee that the performances between them are fair.

In order to alleviate the above problem, in this paper, we propose to build an **unbiased fairness aware recommendation framework** (called UFAR for short). Our general idea is to leverage the inverse propensity score (IPS) [22] to downweight the user-item pairs with higher observational probabilities, and assign larger weights to the underrepresented samples. Based on such weight adjustments, different samples are expected to have fair rights to represent the group performance. While this seems to be an interesting idea, there are still many challenges: to begin with, while IPS has been widely studied in many domains, such as causal inference, computer vision, etc. It is still not clear how to use IPS to debias the fairness. And then, the recommendation datasets are usually noisy and sparse. Directly estimating IPS based on such data can be not reliable, which may lead to sub-optimal recommendation and fairness performances. How to design a more reliable IPS estimation mechanism may also challenge our idea. To overcome these challenges, we firstly define an ideal fairness objective. Then, to make this objective optimizable, we propose an unbiased objective for the observational datasets based on IPS. In our model, we do not expect the IPS directly estimated from the dataset to be absolutely correct. Instead, we assume that the real and predicted IPS can be bounded within a small range. With this assumption, we firstly find the IPS which can lead to the largest loss in the above range, and then minimize this loss to learn the model parameters. By this strategy, we actually optimize the objective based on a range of IPS instead of a specific point, which can better resist the negative impact brought by the incorrect IPS estimation. In addition, to

make our framework more adaptive for Top-N recommendation, we smooth the ranking metrics such as Precision, HR, and NDCG to measure the performance gaps between different user groups, based on which we build an end-to-end framework to achieve unbiased fairness aware Top-N recommendation.

In summary, the main contributions of this paper can be concluded as follows:

- we analyze the potential biases of the recommendation fairness in observation datasets¹, which to the best of our knowledge, is the first time in the recommendation domain.
- To achieve the above goal, we firstly define an ideal fairness-aware learning objective, and then approximate it by an unbiased target for easy optimization. We also improve our model by a range-based IPS strategy and soften the ranking metrics to adapt Top-N recommendation.
- We conduct extensive experiments to demonstrate the effectiveness of our framework.

2 PRELIMINARIES

2.1 User-oriented Fairness

Fairness in the recommendation domain can be item-oriented or user-oriented, where the former cares more about the item exposure opportunities [16], while the latter aims to reduce the performance gaps between different user individuals or user groups [30]. According to the fairness measurement, user-oriented fairness includes individual-level fairness [20] and group-level fairness [10]. Usually, individual-level fairness requires that similar users should be treated similarly [30, 32], while group-level fairness aims to lower the performance discrepancy between different user groups. In this paper, we focus on user-oriented group-level fairness. Formally, suppose we have a user set $\mathcal{U}$ and an item set $\mathcal{V}$, all the interactions of a user $u \in \mathcal{U}$ are collected in a item set $\mathcal{V}_u$. Suppose s is the sensitive feature, indicating different user groups², then group-level fairness requires that the average performance gap between different user groups should be small enough, which is defined as follows:

$$-\epsilon < \frac{1}{|\mathcal{U}_{s=0}|} \sum_{u \in \mathcal{U}_{s=0}} M(u, \mathcal{V}_u) - \frac{1}{|\mathcal{U}_{s=1}|} \sum_{u \in \mathcal{U}_{s=1}} M(u, \mathcal{V}_u) < \epsilon, \quad (1)$$

where $M(u, \mathcal{V}_u)$ is the performance of user u based on her interaction set $\mathcal{V}_u$. Here, we explicitly write out the item sets (i.e., $\mathcal{V}_u$) for computing the performances, since it is the basic cause of the fairness bias. $\mathcal{U}_{s=0}$ and $\mathcal{U}_{s=1}$ are the sets of users whose sensitive features are $s = 0$ and $s = 1$, respectively. ϵ is a small positive value.

2.2 Debaised Recommendation

In the recommendation domain, the data bias has long been a fundamental problem. Basically, the bias is rooted in the user self-selection tendency or the non-uniform item exposure probability in the previous recommender system [1, 25, 28, 36]. In order to remove the bias, recent years have witnessed a large number of

¹It should be noted that our paper does not aim to achieve fairness by debiasing the data. We study the biases of the fairness itself.

²In this paper, we focus on binary sensitive features, but it is not hard to extend our framework to the multi-category sensitive features.

promising models. In general, the key of these studies is to derive unbiased objectives, which are usually in the form of

$$L_{\text{unbiased}} = \frac{1}{|\mathcal{U}||\mathcal{V}|} \sum_{u \in \mathcal{U}} \sum_{v \in \mathcal{V}_u^*} \frac{1}{p(o_{uv} = 1)} l_{uv}(f), \quad (2)$$

where o_{uv} is a binary value, indicating whether the user-item pair (u, v) is observed, $p(o_{uv} = 1)$ is the probability of observing this pair in the dataset. $l_{uv}(f)$ is the loss for (u, v) based on the recommender model f , which can be traditional methods like matrix factorization [14] or deep models such as neural collaborative filtering [9]. $\mathcal{V}_u^*$ is the set of training items for user u . Usually, for rating prediction, $l_{uv}(f)$ is realized by the rooted mean square error (RMSE), and $\mathcal{V}_u^* = \mathcal{V}_u$. For Top-N recommendation, $l_{uv}(f)$ is implemented with the binary cross entropy, and $\mathcal{V}_u^* = \mathcal{V}_u \cup \mathcal{V}_u^-$, where $\mathcal{V}_u^-$ is the set of negative samples. Theoretically, the above L_{unbiased} can be demonstrated to be unbiased in terms of the ideal loss $L_{\text{ideal}} = \frac{1}{|\mathcal{U}||\mathcal{V}|} \sum_{u \in \mathcal{U}} \sum_{v \in \mathcal{V}} l_{uv}(f)$ in expectation [25, 38].

While there are many promising models in the fields of fairness aware recommendation and debaised recommendation, respectively, there is few work on studying the bias of the fairness, which is important to guarantee that the fairness learned on the training set can be consistent on all the items.

3 THE UFAR MODEL

In this section, we firstly define an ideal objective for fairness aware recommendation. Then, to make this objective optimizable, we propose an unbiased loss function based on IPS for easy optimization. To make the loss more reliable, we introduce a range-based strategy to enhance the IPS estimation. At last, we further improve the loss with soft ranking metrics for adapting the Top-N recommendation task, which is more practical for real-world scenarios.

3.1 Ideal Fairness aware Objective

Ideally, a fair recommender model should perform well on all the user-item pairs and provide similar performance for the users in different groups based on all the items. Thus, we have the following ideal optimization problem:

$$\begin{aligned} \min_{\theta_f} \quad & \frac{1}{|\mathcal{U}||\mathcal{V}|} \sum_{u \in \mathcal{U}} \sum_{v \in \mathcal{V}} l_{uv}(f), \\ \text{s.t.} \quad & -\epsilon < \frac{1}{|\mathcal{U}_{s=0}|} \sum_{u \in \mathcal{U}_{s=0}} M(u, \mathcal{V}) - \frac{1}{|\mathcal{U}_{s=1}|} \sum_{u \in \mathcal{U}_{s=1}} M(u, \mathcal{V}) < \epsilon, \end{aligned} \quad (3)$$

where θ_f denotes the parameters of the recommendation model f . The objective optimizes the model for each user-item pair, and the fairness constraint is imposed on the whole item set $\mathcal{V}$. If we compare the above fairness constraint with objective (1), we can see if the observed interaction set $\mathcal{V}_u$ in objective (1) is biased toward many priority items, then the performance gap for the underrepresented items can be less controlled.

3.2 Unbiased Fairness aware Objective

While the ideal objective (3) can lead to the optimal fairness aware recommender model, it is not optimizable, since we cannot access the ground truths for all the user-item pairs. To solve this problem,

we design an unbiased approximator of (3) based on the observational dataset as follows:

$$\begin{aligned} \min_{\theta_f} \quad & \frac{1}{|\mathcal{U}||\mathcal{V}|} \sum_{u \in \mathcal{U}} \sum_{v \in \mathcal{V}_u^*} \frac{1}{p(o_{uv} = 1)} l_{uv}(f), \\ \text{s.t.} \quad & -\epsilon < \frac{1}{N_0} \sum_{(u,v) \in S_0} \frac{M_{u,v}}{p_0(o_{u,v} = 1)} - \frac{1}{N_1} \sum_{(u,v) \in S_1} \frac{M_{u,v}}{p_1(o_{u,v} = 1)} < \epsilon, \end{aligned} \quad (4)$$

where S_0 and S_1 are the sets of the observed user interactions in groups $\mathcal{U}_{s=0}$ and $\mathcal{U}_{s=1}$, respectively. $N_0 = |\mathcal{U}_0||\mathcal{V}|$ and $N_1 = |\mathcal{U}_1||\mathcal{V}|$ are normalization factors. We expand $M(u, \mathcal{V})$ in objective (3) as $M(u, \mathcal{V}) = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} M_{u,v}$, where $M_{u,v}$ is the model prediction accuracy for user u on item v , and in the following, we specify $M_{u,v}$ with the loss function or the soft ranking metrics. p_0 and p_1 are the propensity scores for different user groups, which are computed based on S_0 and S_1 , respectively. The propensity score p can be derived from p_0 and p_1 according to $p(o_{uv} = 1) = p(o_{u,v} = 1, u_s = 0) + p(o_{u,v} = 1, u_s = 1) = p_0(o_{u,v} = 1 | u_s = 0) p(u_s = 0) + p_1(o_{u,v} = 1 | u_s = 1) p(u_s = 1)$, where u_s represents the sensitive feature of user u . Since each user can only belong to one user group, we have $p(o_{u,v} = 1) = p_0(o_{u,v} = 1)$ and $p(o_{u,v} = 1) = p_1(o_{u,v} = 1)$ for the users in group $\mathcal{U}_{s=0}$ and $\mathcal{U}_{s=1}$, respectively. For the unbiasedness of objective (4), we have the following theory:

Theorem 1. Let $L_1 = \frac{1}{|\mathcal{U}||\mathcal{V}|} \sum_{u \in \mathcal{U}} \sum_{v \in \mathcal{V}_u^*} \frac{1}{p(o_{uv} = 1)} l_{uv}(f)$ and $L_2 = \frac{1}{N_0} \sum_{(u,v) \in S_0} \frac{M_{u,v}}{p_0(o_{u,v} = 1)} - \frac{1}{N_1} \sum_{(u,v) \in S_1} \frac{M_{u,v}}{p_1(o_{u,v} = 1)}$, then L_1 and L_2 are unbiased in terms of the objective and constraint in problem (3), respectively, that is:

$$\begin{aligned} \mathbb{E}[L_1] &= \frac{1}{|\mathcal{U}||\mathcal{V}|} \sum_{u \in \mathcal{U}} \sum_{v \in \mathcal{V}} l_{uv}(f), \\ \mathbb{E}[L_2] &= \frac{1}{|\mathcal{U}_{s=0}|} \sum_{u \in \mathcal{U}_{s=0}} M(u, \mathcal{V}) - \frac{1}{|\mathcal{U}_{s=1}|} \sum_{u \in \mathcal{U}_{s=1}} M(u, \mathcal{V}). \end{aligned} \quad (5)$$

PROOF. For L_1 , we have:

$$\begin{aligned} \mathbb{E}[L_1] &= \mathbb{E}\left[\frac{1}{|\mathcal{U}||\mathcal{V}|} \sum_{u \in \mathcal{U}} \sum_{v \in \mathcal{V}_u^*} \frac{1}{p(o_{uv} = 1)} l_{uv}(f)\right] \\ &= \frac{1}{|\mathcal{U}||\mathcal{V}|} \sum_{u \in \mathcal{U}} \sum_{v \in \mathcal{V}} \frac{\mathbb{E}[o_{uv}]}{p(o_{uv} = 1)} l_{uv}(f) \\ &= \frac{1}{|\mathcal{U}||\mathcal{V}|} \sum_{u \in \mathcal{U}} \sum_{v \in \mathcal{V}} l_{uv}(f), \end{aligned}$$

where the third line holds because $\mathbb{E}[o_{uv}] = 1 * p(o_{uv} = 1) + 0 * p(o_{uv} = 0) = p(o_{uv} = 1)$. For L_2 , we have:

$$\begin{aligned} \mathbb{E}[L_2] &= \mathbb{E}\left[\frac{1}{N_0} \sum_{(u,v) \in S_0} \frac{M_{u,v}}{p_0(o_{u,v} = 1)} - \frac{1}{N_1} \sum_{(u,v) \in S_1} \frac{M_{u,v}}{p_1(o_{u,v} = 1)}\right] \\ &= \mathbb{E}\left[\frac{1}{N_0} \sum_{u \in \mathcal{U}_{s=0}} \sum_{v \in \mathcal{V}_u} \frac{M_{u,v}}{p_0(o_{u,v} = 1)} - \frac{1}{N_1} \sum_{u \in \mathcal{U}_{s=1}} \sum_{v \in \mathcal{V}_u} \frac{M_{u,v}}{p_1(o_{u,v} = 1)}\right] \\ &= \mathbb{E}\left[\frac{1}{N_0} \sum_{u \in \mathcal{U}_{s=0}} \sum_{v \in \mathcal{V}} \frac{o_{u,v} M_{u,v}}{p_0(o_{u,v} = 1)} - \frac{1}{N_1} \sum_{u \in \mathcal{U}_{s=1}} \sum_{v \in \mathcal{V}} \frac{o_{u,v} M_{u,v}}{p_1(o_{u,v} = 1)}\right] \\ &= \frac{1}{N_0} \sum_{u \in \mathcal{U}_{s=0}} \sum_{v \in \mathcal{V}} \frac{\mathbb{E}[o_{u,v}] M_{u,v}}{p_0(o_{u,v} = 1)} - \frac{1}{N_1} \sum_{u \in \mathcal{U}_{s=1}} \sum_{v \in \mathcal{V}} \frac{\mathbb{E}[o_{u,v}] M_{u,v}}{p_1(o_{u,v} = 1)} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{N_0} \sum_{u \in \mathcal{U}_{s=0}} \sum_{v \in \mathcal{V}} M_{u,v} - \frac{1}{N_1} \sum_{u \in \mathcal{U}_{s=1}} \sum_{v \in \mathcal{V}} M_{u,v} \\
&= \frac{1}{N_0} \sum_{u \in \mathcal{U}_{s=0}} |\mathcal{V}| M(u, \mathcal{V}) - \frac{1}{N_1} \sum_{u \in \mathcal{U}_{s=1}} |\mathcal{V}| M(u, \mathcal{V}) \\
&= \frac{1}{|\mathcal{U}_{s=0}|} \sum_{u \in \mathcal{U}_{s=0}} M(u, \mathcal{V}) - \frac{1}{|\mathcal{U}_{s=1}|} \sum_{u \in \mathcal{U}_{s=1}} M(u, \mathcal{V}).
\end{aligned}$$

3.3 Range-based IPS

Usually, the IPS is estimated from observational datasets [26]. In objective (4), suppose $\hat{p}_0$ and $\hat{p}_1$ are the predicted values for p_0 and p_1 , respectively, then we have the following objective:

$$\begin{aligned}
&\min_{\theta_f} \frac{1}{|\mathcal{U}| |\mathcal{V}|} \sum_{u \in \mathcal{U}} \sum_{v \in \mathcal{V}_u^*} \frac{1}{\hat{p}(o_{uv}=1)} l_{uv}(f), \\
&\text{s.t. } -\epsilon < \frac{1}{N_0} \sum_{(u,v) \in S_0} \frac{M_{u,v}}{\hat{p}_0(o_{uv}=1)} - \frac{1}{N_1} \sum_{(u,v) \in S_1} \frac{M_{u,v}}{\hat{p}_1(o_{uv}=1)} < \epsilon,
\end{aligned} \quad (6)$$

where $\hat{p}(o_{uv}=1) = \hat{p}_0(o_{uv}=1)$, if $(u,v) \in S_0$, otherwise, $\hat{p}(o_{uv}=1) = \hat{p}_1(o_{uv}=1)$.

In practice, the recommendation datasets can be quite sparse and noisy, which may greatly impact the accuracy of IPS estimation. To alleviate this problem, we introduce a range-based IPS strategy for model optimization. In general, we firstly estimate an initial IPS directly from the training data, and then we assume that the real IPS is within a δ -ball centered at the initial IPS. At last, we learn our model by minimizing the maximum loss induced by the IPS in this δ -ball range. Formally, we have the following objective:

$$\begin{aligned}
&\min_{\theta_f} \max_{\theta_p} \frac{1}{|\mathcal{U}| |\mathcal{V}|} \sum_{u \in \mathcal{U}} \sum_{v \in \mathcal{V}_u^*} \frac{1}{\bar{p}(o_{uv}=1)} l_{uv}(f), \\
&\text{s.t. } -\epsilon < \frac{1}{N_0} \sum_{(u,v) \in S_0} \frac{M_{u,v}}{\bar{p}_0(o_{uv}=1)} - \frac{1}{N_1} \sum_{(u,v) \in S_1} \frac{M_{u,v}}{\bar{p}_1(o_{uv}=1)} < \epsilon, \quad (7) \\
&|\bar{p}_0(o_{uv}=1) - \hat{p}_0(o_{uv}=1)| < \delta, \quad \forall (u,v) \in S_0, \\
&|\bar{p}_1(o_{uv}=1) - \hat{p}_1(o_{uv}=1)| < \delta, \quad \forall (u,v) \in S_1,
\end{aligned}$$

where we use $\bar{p}_0(o_{uv}=1)$ and $\bar{p}_1(o_{uv}=1)$ to learn the IPS that can lead to the largest loss. We implement $\bar{p}_0(o_{uv}=1)$ and $\bar{p}_1(o_{uv}=1)$ with three-layer fully connected neural networks, and their parameters are collected in θ_p . $\bar{p}(o_{uv}=1)$ is either $\bar{p}_0(o_{uv}=1)$ or $\bar{p}_1(o_{uv}=1)$ according to the sensitive feature of u . δ is a positive value. Basically, objective (7) is a minimax game. In the “max” operation, we search the IPS in the δ -ball which can lead to the largest loss. The obtained result can be seen as the “boundary” IPS, and if our model can perform well with this worst estimated IPS, then we can reasonably infer that the loss with the real IPS can be not large. As a result, in the “min” operation, we minimize the loss with the “boundary” IPS.

3.4 Ranking based Fairness Constraint

Previous works usually build fairness constraints based on the rating prediction performance [37]. However, the recommendation task is basically a ranking problem. To make the fairness constraint more adaptive for the ranking problem, we propose the following two methods to specify $M_{u,v}$ in objective (7).

(i) We use the loss function as an approximation of the ranking based performance, and we let $M_{u,v} = l_{uv}(f)$.

(ii) To directly optimize the ranking metrics and also keep the framework differentiable, we build the fairness constraint based on the soft ranking strategy [6]. Formally, suppose the preference scores predicted for user u on all the items are $\mathbf{s}_u = [s_{u1}, \dots, s_{u|\mathcal{V}|}]$, then the probability that item v is ranked at position k in the recommendation list:

$$p_{kv}^u = [\text{SOFTMAX} \{((N+1-2k)\mathbf{s}_u - \mathbf{A} \cdot \mathbf{1}) / \tau\}]_v, \quad (8)$$

where $\mathbf{A} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$, and $A_{ij} = |s_{ui} - s_{uj}|$. $\mathbf{1} \in \mathbb{R}^{|\mathcal{V}|}$ is an all one column vector. τ is a temperature parameter. “SOFTMAX” is the softmax operator. $[\cdot]_v$ is the v th element of a vector. For the complete derivation process of p_{kv}^u , we refer the readers to [6] for more details. Based on p_{kv}^u , the ranking metrics like Precision, Hit-Ratio, and NDCG [15, 29] can be softened. In our model, we use Precision@K for implementation, and for a user u , the softened Precision@K is:

$$\text{Precision@K} = \frac{1}{K} \sum_{k=1}^K \sum_{v \in \mathcal{V}_u} p_{kv}^u, \quad (9)$$

where K is the number of recommended items. Intuitively, if the user actually interacted items (i.e., $\mathcal{V}_u$) are predicted to be ranked within the first K positions with higher probabilities, then the performance of the recommendation model is better, and correspondingly, the Precision@K is larger.

To relate the above Precision@K with $M_{u,v}$, we rewrite it as follows:

$$\text{Precision@K} = \sum_{v \in \mathcal{V}_u} \frac{1}{K} \sum_{k=1}^K p_{kv}^u, \quad (10)$$

where $\frac{1}{K} \sum_{k=1}^K p_{kv}^u$ can be seen as the model performance on item v . Larger $\frac{1}{K} \sum_{k=1}^K p_{kv}^u$ means that, for a ground truth item v , the model is more likely to successfully recommend v (i.e., item v is ranked within the first K positions). Inspired by this analysis, we specify $M_{u,v}$ with $\frac{1}{K} \sum_{k=1}^K p_{kv}^u$. Since p_{kv}^u is directly associated with the model outputs, the fairness constraint in objective (7) is fully differentiable and can be easily optimized with various gradient descent algorithms.

3.5 Overall Objective

By combining the above designs, and inspired by the method of Lagrange multipliers [39], our final objective is:

$$\begin{aligned}
L_{\text{final}}(\theta_f, \theta_p) &= \frac{1}{|\mathcal{U}| |\mathcal{V}|} \sum_{u \in \mathcal{U}} \sum_{v \in \mathcal{V}_u^*} \frac{1}{\bar{p}(o_{uv}=1)} l_{uv}(f) \\
&+ \lambda \left| \frac{1}{N_0} \sum_{(u,v) \in S_0} \frac{M_{u,v}}{\bar{p}_0(o_{uv}=1)} - \frac{1}{N_1} \sum_{(u,v) \in S_1} \frac{M_{u,v}}{\bar{p}_1(o_{uv}=1)} \right| \\
&+ \gamma \sum_{S_0} |\bar{p}_0(o_{uv}=1) - \hat{p}_0(o_{uv}=1)| \\
&+ \gamma \sum_{S_1} |\bar{p}_1(o_{uv}=1) - \hat{p}_1(o_{uv}=1)|,
\end{aligned} \quad (11)$$

Algorithm 1: Learning Algorithm of UFAR

Input: The number of training epochs T ; The number of batches B ; The learning rate α_p and α_f ; The number of iterations N_p and N_f .

```

begin
  Obtain the initial propensity scores  $\hat{p}_0$  and  $\hat{p}_1$ .
  for  $i$  in  $[0, T]$  do
    for  $j$  in  $[0, B]$  do
      Sample a mini-batch from the training data.
      for  $k_1$  in  $[0, N_p]$  do
         $\theta_p \leftarrow \theta_p + \alpha_p \frac{\partial L_{\text{final}}(\theta_f, \theta_p)}{\partial \theta_p}$ .
      end
      for  $k_2$  in  $[0, N_f]$  do
         $\theta_f \leftarrow \theta_f - \alpha_f \frac{\partial L_{\text{final}}(\theta_f, \theta_p)}{\partial \theta_f}$ .
      end
    end
  end
end
end

```

where λ and γ are weighting parameters. $M_{u,v}$ can be implemented based on the loss or the softened ranking metrics. In this paper, we use cross entropy (CE) to implement l_{uv} for Top-N recommendation.

The complete training process of our framework is shown in Algorithm 1. To begin with, we obtain the initial propensity scores $\hat{p}_0$ and $\hat{p}_1$ based on the common methods like [22, 25, 35]. Then for each training batch, we iteratively update θ_p and θ_f for N_p and N_f times to maximize and minimize $L_{\text{final}}(\theta_f, \theta_p)$, respectively. At last, the model θ_f is leveraged to provide fair and accurate recommendations.

4 RELATED WORK

Our method is related with the fields of fairness aware recommendation and debiased recommendation. In the following, we briefly review these fields, and discuss the relation between our paper and the previous work.

4.1 Fairness Aware Recommendation

Fairness aware recommendation can be classified into item-oriented and user-oriented methods. In the item-oriented methods, the model is usually optimized to provide equal exposure opportunities for different items [16]. In the user-oriented methods, people have defined individual-level fairness and group-level fairness. In this paper, our focus is the user-oriented group-level fairness. In the past few decades, there are a lot of promising models proposed along this line. For example, [37] proposes four types of fairness constraints on the matrix factorization model. [17] designs a model to remove the mainstream bias to achieve the recommendation fairness. [32] leverages adversarial training to learn the user embeddings which cannot predict sensitive information. [19] defines personalized counterfactual fairness based on causal notions, and builds an adversarial model to achieve it. [18] provides a post-processing strategy to achieve fairness aware recommendation by

solving an integer programming problem. While the above models have achieved many successes, they mostly focus on the models. However, in our paper, we consider fairness from the data perspective and build an unbiased fairness aware recommender framework, which significantly differs from the previous work.

4.2 Debiased recommendation

To overcome the data bias problem in the recommendation, people have proposed a large amount of debiased recommender models. For example, [25] uses propensity score to reweigh different training samples for building an unbiased learning objective, which is demonstrated to be quite effective for correcting the data bias. To remove the bias of the user implicit feedback, [24] proposes an IPS strategy to revise the cross entropy loss. [22] proposes an asymmetric tri-training framework based on two predictors to correct the data bias. [3] proposes a framework to achieve robust learning based on the reweighing mechanism. [7] proposes a doubly robust estimator to alleviate the high variance issues brought by the traditional imputation models. Most of the above methods aim to improve the overall model performance by correcting the biases of the user-item interactions. However, we care more about the fairness problem between different user groups. To the best of our knowledge, this is the first work on debiased fairness aware recommendation.

5 EXPERIMENTS

In this section, we conduct extensive experiments to demonstrate the effectiveness of our framework, where we focus on the following research questions:

RQ1: whether our framework can be more effective than the other methods for different base models?

RQ2: How do different components in our framework contribute the final performance?

RQ3: How does the data bias level influence the final performance of our framework?

RQ4: Whether our framework can lead to more fair performance on the underrepresented items?

RQ5: How do different hyper-parameters in our framework influence the final performance?

In the following, we first introduce the experiment setup, and then answer the above questions by presenting and analyzing the empirical results.

5.1 Experiment Setup

Datasets. We conduct our experiments based on the following real-world datasets:

- **Yahoo! R3**³ is a dataset containing the user explicit feedback on the music, which is collected from 2002 to 2006.

- **Coat**⁴ is a dataset about user preferences on the clothes. For each of these datasets, the training set is biased, which is collected based on the normal user-item interactions. The items in the testing sets are uniformly exposed to the users, which makes the evaluation fairly consider the priority and underrepresented items for different user groups. The user feedback is integer ratings falling between 1

³<http://webscope.sandbox.yahoo.com/>

⁴<https://www.cs.cornell.edu/~schnabts/mnar/>

Table 1: Statistics of the datasets used in our experiments

Dataset	Coat	Yahoo	ML-1M	WeChat
# User	290	15,400	6,040	2,879
# Item	300	1,000	3,706	1,722
# Interaction	7,820	316,453	1,247,368	715,770
Sparsity	91.07%	97.95%	94.43%	85.58%

and 5. To adapt Top-N recommendation, we regard the interaction as positive if the rating is larger than 3, while the other interactions are seen as negative samples.

In addition to the above unbiased datasets, we also conduct experiments on the following recommendation datasets:

- **MovieLens (ML)-1M⁵** and **WeChat⁶**, which contain user preferences on the movies and short videos, respectively.

However, these datasets are biased, and to fairly evaluate different items, we follow the common practice [7, 23, 24] to build the unbiased versions of these datasets. More specifically,

(a) we firstly use matrix factorization (MF) [14] to complete the user-item rating matrix, and the obtained rating for a user-item pair (u, v) is $\hat{r}_{u,v}$.

(b) Then, we construct an observation matrix, where if a user-item pair is observed in the dataset, then the corresponding matrix value is 1, otherwise, the value is 0. We use logistic matrix factorization [11] to complete this observation matrix, and the estimated observation for a user-item pair (u, v) is $\hat{o}_{u,v}$.

(c) Next, we compute the user-item relevance and observation parameters, which are denoted as $\gamma_{u,v} = \sigma(\hat{r}_{u,v} - \zeta)$, $\phi_{u,v} = (\hat{o}_{u,v})^\beta$, $\forall u \in \mathcal{U}, v \in \mathcal{V}$, where σ is the sigmoid function. ζ is a constant indicating the average positive level of the dataset. A smaller ζ will lead to more positive interactions. β is a constant controlling the bias level of the dataset. A larger β will make the dataset more biased.

(d) Finally, we generate the unbiased dataset as follows:

$$r_{u,v} \sim B(\gamma_{u,v}), \quad o_{u,v} \sim B(\phi_{u,v}), \quad \forall u \in \mathcal{U}, v \in \mathcal{V},$$

where B is the Bernoulli distribution.

After obtaining the unbiased datasets, we randomly select 5% user-item interactions as the testing sets, while the others are left for training. In our experiments, to explore different types of sensitive variables, we group different users according to their activity levels or genders. For Yahoo, ML-1M and WeChat, we rank different users according to their interaction numbers, and the top 5% users are clustered into one group (called advantaged group), while the others are regarded as another group (called disadvantaged group). For Coat, we group different users according to their genders. The statistics of the above datasets are presented in Table 1.

Baselines. To evaluate whether our framework can be generally effective, we specify the recommender model f in objective (11) with the following methods:

- **MF** [14] is a well known matrix factorization model, which is a shadow architecture and focuses on the linear user-item relations.

- **NCF** [9] is a deep recommender model, which can capture the non-linear correlations between the users and items.

- **LightGCN** [8] is a graph based recommender model, which enhances the collaborative filtering information by the user-item interaction structures. For each of the above recommender models, we compare our framework with the following debiased and fairness aware models:

- **RT** [25] is a famous debiased recommendation model based on IPS, where the rating prediction module is the matrix factorization model.

- **Implicit** [24] is also an unbiased model, but it aims to model the user implicit feedback.

- **MRDR** [3] is a doubly robust method, where the imputation model is improved to lower the objective variance.

- **DRBIAS** [2] is also a doubly robust model, which has smaller bias through optimizing the imputation model.

- **FOCF** [37] is a commonly used group-level fairness aware recommender model, where the fairness constraint is built based on the rating prediction performance.

- **UFAR** is a variant of our model, where we do not correct the data bias (i.e., we remove the IPS from objective (11)).

- **UFAR-L** and **UFAR** are both our models, where the performance $M_{u,v}$ is implemented based on the loss function and the soft ranking metric, respectively.

It should be noted that our framework cannot be directly compared with the post-processing models like [18, 21], since they need to be executed based on the ground truth testing set. Because we focus on group-level fairness, the adversarial learning based models [19, 32] which are mostly leveraged for individual fairness are also not comparable.

Implementation details. We evaluate different models on the recommendation performance and user group-level fairness. For the recommendation performance, the commonly used metrics including **NDCG@10** and **F1@10** are leveraged in the experiments [4, 5, 33, 41], and we report the results based on different user groups and the complete datasets. For measuring the group-level fairness, we follow the previous work [10, 18, 37, 40] to evaluate the performance gap (**PG**) between different user groups. In specific, suppose $\mathcal{D}_{s=0}$ and $\mathcal{D}_{s=1}$ are the sets of users with different sensitive features in the testing set, then we have:

$$\text{PG} = \left| \frac{1}{|\mathcal{D}_{s=0}|} \sum_{u \in \mathcal{D}_{s=0}} s_u - \frac{1}{|\mathcal{D}_{s=1}|} \sum_{u \in \mathcal{D}_{s=1}} s_u \right|,$$

where s_u is the performance of user u in terms of NDCG@10 or F1@10. The parameters of our model are tuned based on grid search. More specifically, the learning rate and parameter decaying ratio (i.e., weight decay) are tuned in the range of $\{0.1, 0.01, 0.001, 0.0001\}$ and $\{0, 1e-6, 1e-5, 1e-4, 1e-3\}$, respectively. The Lagrange coefficients λ and γ are searched in the range of $\{0.01, 0.1, 1, 10, 100\}$. In the soft ranking metrics, we set $K = 10$, and tune τ in the range of $\{0.1, 1, 2, 4, 8, 16\}$. The parameters of the baselines are tuned in the same ranges as our model's. For all the models, we use Adam [13] for parameter optimization.

⁵<https://grouplens.org/datasets/movielens/>

⁶https://github.com/WeChat-Big-Data-Challenge-2021/WeChat_Big_Data_Challenge

Table 2: Overall comparison between our model and the baselines. The performances are reported based on different user groups as well as the complete dataset. We use bold fonts to label the best overall recommendation and fairness performances. The results are percentage value with “%” omitted. “↑” or “↓” indicates that the performance is better if the reported value is larger or smaller. The improvements of our model against the best baseline is significant under paired-t test with $p < 0.05$.

		NDCG@10 (Yahoo)				F1@10 (Yahoo)				NDCG@10 (Coat)				F1@10 (Coat)			
		Disadv.	Adv.	Overall↑	PG↓	Disadv.	Adv.	Overall↑	PG↓	Disadv.	Adv.	Overall↑	PG↓	Disadv.	Adv.	Overall↑	PG↓
MF	Orig.	4.044	2.727	2.860	1.317	2.047	1.250	1.331	0.797	12.88	18.14	15.78	5.265	5.095	7.890	6.655	2.795
	RT	3.844	2.729	2.842	1.115	1.981	1.261	1.334	0.720	13.92	17.69	16.00	3.770	4.866	7.597	6.383	2.731
	Implicit	4.283	2.796	2.946	1.487	2.212	1.386	1.470	0.826	14.70	17.60	16.30	2.894	5.165	7.492	6.490	2.327
	MRDR	2.586	1.565	1.668	1.021	1.348	0.691	0.757	0.657	14.40	17.01	15.83	2.608	5.495	7.578	6.658	2.083
	DRBIAS	3.304	1.809	1.960	1.495	1.672	0.655	0.760	1.017	13.75	16.58	15.31	2.835	5.577	7.737	6.770	2.160
	FOCF	4.309	2.760	2.916	1.549	2.125	1.184	1.279	0.941	12.28	17.98	15.42	5.703	4.887	7.909	6.570	3.022
	UFAR-	3.791	1.839	2.036	1.952	1.444	0.815	0.878	0.629	13.51	16.60	15.21	3.093	5.070	7.254	6.302	2.184
	UFAR-L	4.469	2.698	2.876	1.771	1.803	1.163	1.228	0.640	13.77	17.12	15.62	3.346	5.334	7.925	6.802	2.591
	UFAR	3.643	2.870	2.947	0.773	1.866	1.303	1.361	0.563	15.22	17.20	16.31	1.986	6.607	8.173	7.504	1.566
NCF	Orig.	3.492	1.999	2.150	1.493	1.281	0.884	0.926	0.397	15.77	17.20	16.56	1.431	5.930	8.134	7.173	2.204
	RT	3.176	1.984	2.104	1.192	1.449	0.948	0.998	0.501	14.79	19.92	17.62	5.133	5.135	8.891	7.231	3.756
	Implicit	1.224	2.283	2.176	1.059	0.581	1.000	0.959	0.421	15.49	17.75	16.73	2.265	4.834	7.582	6.357	2.748
	MRDR	2.959	2.171	2.250	0.788	1.421	0.990	1.034	0.431	14.34	18.42	16.58	4.075	5.070	7.690	6.533	2.620
	DRBIAS	2.883	2.198	2.267	0.685	1.361	0.977	1.016	0.384	15.40	17.64	16.63	2.243	5.749	8.191	7.122	2.442
	FOCF	3.552	2.080	2.229	1.472	1.278	0.923	0.958	0.355	15.98	17.90	17.04	1.917	5.759	8.054	7.070	2.295
	UFAR-	2.795	1.809	1.908	0.986	1.190	0.837	0.873	0.353	15.58	16.66	16.18	1.076	4.902	7.526	6.373	2.624
	UFAR-L	3.105	1.882	2.006	1.223	1.640	0.917	0.989	0.723	16.55	18.35	17.54	1.798	5.100	7.341	6.360	2.241
	UFAR	2.460	2.323	2.337	0.137	1.349	0.999	1.035	0.350	17.87	17.96	17.92	0.089	6.411	8.145	7.406	1.734
LGC	Orig.	4.647	3.712	3.806	0.935	2.141	1.691	1.736	0.450	16.30	18.54	17.53	2.240	5.140	7.715	6.589	2.575
	RT	4.302	3.815	3.864	0.487	1.946	1.613	1.647	0.333	16.42	18.53	17.58	2.105	4.944	7.355	6.289	2.411
	Implicit	4.494	3.849	3.914	0.645	1.961	1.694	1.720	0.267	16.13	18.87	17.64	2.737	5.315	7.776	6.705	2.461
	MRDR	2.624	3.737	3.625	1.113	1.008	1.536	1.482	0.528	15.58	16.56	16.12	0.986	5.048	7.481	6.408	2.433
	DRBIAS	2.624	3.737	3.625	1.113	1.008	1.536	1.482	0.528	15.47	16.26	15.91	0.794	5.292	7.154	6.365	1.862
	FOCF	4.225	3.788	3.832	0.437	2.041	1.633	1.675	0.408	15.77	16.94	16.42	1.169	5.531	7.451	6.606	1.920
	UFAR-	4.257	3.178	3.287	1.079	1.714	1.454	1.480	0.260	15.31	17.00	16.24	1.691	5.214	7.300	6.378	2.086
	UFAR-L	4.220	3.518	3.589	0.702	1.976	1.511	1.557	0.465	15.82	19.41	17.79	3.588	5.259	7.657	6.599	2.398
	UFAR	4.004	4.044	4.040	0.040	1.965	1.774	1.794	0.191	17.74	18.16	17.98	0.421	5.813	7.359	6.706	1.546
		NDCG@10 (WeChat)				F1@10 (WeChat)				NDCG@10 (ML-1M)				F1@10 (ML-1M)			
		Disadv.	Adv.	Overall↑	PG↓	Disadv.	Adv.	Overall↑	PG↓	Disadv.	Adv.	Overall↑	PG↓	Disadv.	Adv.	Overall↑	PG↓
MF	Orig.	1.832	1.348	1.372	0.484	1.338	0.911	0.933	0.427	1.922	1.326	1.356	0.596	1.046	0.811	0.823	0.235
	RT	1.984	1.361	1.392	0.623	1.496	0.903	0.934	0.593	2.027	1.351	1.384	0.676	1.117	0.819	0.834	0.298
	Implicit	0.815	1.423	1.393	0.608	0.589	0.980	0.960	0.391	1.775	1.364	1.384	0.411	1.147	0.855	0.869	0.292
	MRDR	0.719	1.402	1.368	0.683	0.525	0.915	0.897	0.390	1.831	1.269	1.297	0.562	1.137	0.800	0.818	0.337
	DRBIAS	0.740	1.400	1.367	0.660	0.536	0.905	0.886	0.369	1.727	1.299	1.320	0.428	0.991	0.785	0.795	0.206
	FOCF	1.588	1.329	1.342	0.259	1.234	0.854	0.872	0.380	1.854	1.362	1.386	0.492	1.010	0.847	0.855	0.163
	UFAR-	1.688	1.266	1.287	0.422	1.257	0.866	0.887	0.391	0.953	1.320	1.302	0.367	0.540	0.818	0.804	0.276
	UFAR-L	0.731	1.376	1.343	0.645	0.598	0.935	0.918	0.337	1.829	1.284	1.311	0.545	0.951	0.775	0.784	0.178
	UFAR	1.600	1.387	1.398	0.213	1.212	0.931	0.946	0.281	1.651	1.375	1.389	0.276	1.019	0.875	0.882	0.144
NCF	Orig.	1.485	1.305	1.314	0.180	0.926	0.868	0.872	0.058	1.937	1.379	1.407	0.558	1.103	0.865	0.878	0.238
	RT	1.567	1.342	1.353	0.225	1.236	0.878	0.896	0.358	2.169	1.379	1.419	0.790	1.203	0.878	0.894	0.325
	Implicit	0.827	1.375	1.347	0.548	0.829	0.935	0.929	0.106	1.578	1.249	1.266	0.329	0.908	0.760	0.767	0.148
	MRDR	1.050	1.330	1.316	0.280	0.819	0.917	0.912	0.098	1.854	1.358	1.383	0.496	1.117	0.853	0.866	0.264
	DRBIAS	1.236	1.337	1.332	0.101	1.022	0.913	0.919	0.109	1.623	1.325	1.340	0.298	1.054	0.836	0.847	0.218
	FOCF	1.076	1.366	1.352	0.290	1.113	0.927	0.937	0.186	1.890	1.394	1.419	0.496	1.099	0.884	0.895	0.215
	UFAR-	0.878	1.320	1.298	0.442	0.663	0.841	0.833	0.178	1.984	1.345	1.377	0.639	1.022	0.802	0.814	0.220
	UFAR-L	1.035	1.339	1.324	0.304	0.732	0.910	0.902	0.178	1.842	1.379	1.402	0.463	1.123	0.881	0.893	0.242
	UFAR	1.419	1.493	1.489	0.074	1.030	0.983	0.985	0.047	1.183	1.445	1.432	0.262	0.704	0.913	0.903	0.209
LGC	Orig.	0.920	1.384	1.361	0.464	0.807	0.944	0.937	0.137	1.868	1.352	1.378	0.516	1.195	0.856	0.873	0.339
	RT	0.765	1.373	1.343	0.608	0.756	0.882	0.876	0.126	1.886	1.369	1.395	0.517	1.233	0.862	0.880	0.371
	Implicit	1.720	1.395	1.412	0.325	1.363	0.917	0.939	0.446	1.880	1.368	1.394	0.512	1.173	0.856	0.873	0.317
	MRDR	1.086	1.457	1.438	0.371	0.942	0.962	0.962	0.020	1.676	1.319	1.336	0.357	1.050	0.815	0.827	0.235
	DRBIAS	1.178	1.368	1.358	0.190	0.748	0.887	0.881	0.139	1.815	1.330	1.354	0.485	1.193	0.864	0.880	0.329
	FOCF	1.008	1.446	1.424	0.438	0.930	0.967	0.966	0.037	1.708	1.367	1.384	0.341	1.101	0.862	0.874	0.239
	UFAR-	1.937	1.407	1.433	0.530	1.258	0.935	0.953	0.323	1.849	1.371	1.395	0.478	1.176	0.870	0.886	0.306
	UFAR-L	1.542	1.336	1.346	0.206	1.162	0.917	0.930	0.245	1.787	1.357	1.379	0.430	1.117	0.845	0.858	0.272
	UFAR	1.302	1.480	1.471	0.178	1.003	0.984	0.986	0.019	1.677	1.381	1.396	0.296	1.113	0.882	0.894	0.231

5.2 Overall Comparison

The overall comparison results can be seen in Table 2, from which we can see: compared with the original models, the best debaised

method can usually achieve better overall recommendation performance, which agrees with the previous work [2, 3, 24, 34]. The

reason can be that the original models are optimized based on the biased training set, without debiasing strategies, they cannot perform well on the unbiased testing sets.

While FOCF and UFAR- both have fairness constraints, we find that their fairness performances are not satisfied. In many cases, their performances on PG are even worse than the original model's. We speculate that in FOCF, the fairness is evaluated based on the rating prediction performance, which is not adaptive for the ranking metrics in PG. For UFAR-, the fairness is computed based on the performance of the observational biased dataset, whose distribution differs from the testing unbiased dataset. As a result, both of their performances are not high.

Encouragingly, our framework can obtain the best performance on all the metrics, and the results are consistent on almost all the datasets and original models. Between different implementations of the fairness constraint, the soft ranking strategy is better because it is more aligned with the evaluation metrics. More specifically, compared with the original model, our framework can improve the performance by about 5.070%, 5.596%, 65.92%, and 35.50% on NDCG@10, F1@10, PG(NDCG@10) and PG(F1@10), respectively. Actually, the results are not surprising. The purely debiased models cannot regularize the performance gaps between different user groups, and the IPS in these models is directly estimated from the noisy recommendation datasets. However, in our framework, we explicitly regularize the performance gap and introduce a range-based strategy to predict more reliable IPS, which helps to achieve better performance. Compared with the fairness aware baselines, our framework can simultaneously correct the biases in the recommendation model and fairness constraint, which leads to better performances on different metrics.

If we observe the recommendation performances on different user groups, we find that on Yahoo, ML-1M, and WeChat, the performances of the disadvantaged users can be significantly improved by slightly sacrificing the advantaged user performances. For the Coat dataset, the performance of the male group is enhanced, while the female user performances are lowered. It seems that, for all the datasets, the performance improvement on one group is usually larger than the damage on the other group performance. Thus, our framework can improve the overall performance in most cases.

5.3 Ablation Studies

Our framework is composed of many components. A natural question is whether these components are all necessary, and if yes, how do they contribute to the final performance? To answer this question, we conduct many ablation studies. In specific, we compare our framework with its four variants: in UFAR (w/o F), we drop the fairness constraints in objective (11). In UFAR (w/o L), we drop $\bar{p}_0(o_{u,v} = 1)$ and $\bar{p}_1(o_{u,v} = 1)$, and only use the initially estimated IPS to correct the bias. In UFAR (w/o I), we do not estimate the initial IPS, and only learn $\bar{p}_0(o_{u,v} = 1)$ and $\bar{p}_1(o_{u,v} = 1)$ without any range based constraints. In UFAR (RF), we build fairness constraint based on the rating prediction performance. By UFAR (w/o F), we aim to evaluate whether the fairness constraint in our framework is effective. By UFAR (w/o I) and UFAR (w/o L), we would like to study whether the designed range-based IPS estimation strategy is useful. By UFAR (RF), we hope to investigate whether the ranking based

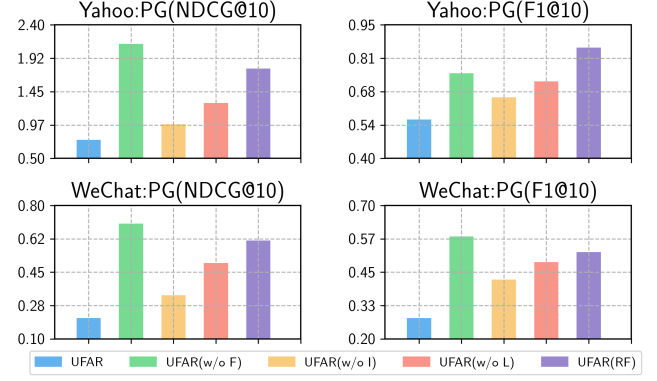


Figure 2: Results of Ablation studies. The results are percentage value with “%” omitted.

fairness constraint is effective. Since the main focus of our paper is improving the recommendation fairness, we evaluate different models based on PG in terms of NDCG@10 and F1@10, respectively. We use the soft ranking to implement the fairness constraint due to its higher performance. The experiment is conducted based on the datasets of Yahoo and WeChat, and we use MF as the original model. The conclusions on the other datasets and original models are similar and omitted.

We present the results in Figure 2, where we can see: UFAR (w/o F) performs much worse than UFAR on all the datasets and evaluation metrics, which is as expected, and demonstrates that the fairness constraint is indeed necessary to lower the performance gap between different user groups. If we drop $\bar{p}_0$ and $\bar{p}_1$, the performance of UFAR (w/o L) is not satisfying compared with UFAR. The reason can be that by learning $\bar{p}_0$ and $\bar{p}_1$, we optimize the “worst-case” loss induced by the IPS in a set. Once the real IPS falls into this set, our framework ensures that the lower bound performance is not worse. If we only optimize the model based on a specific IPS, then performance cannot be guaranteed once the IPS is not accurate. As a result, we observe that UFAR can exhibit better performance. However, only using $\bar{p}_0$ and $\bar{p}_1$ is still sub-optimal, which is evidenced by the lowered performance of UFAR (w/o I) as compared with UFAR. We speculate that blindly learning $\bar{p}_0$ and $\bar{p}_1$ can be too ambitious to efficiently discover the true IPS. If we introduce a prior to guide their searching scope, then the results can be better. In our framework, we use the initially estimated IPS as the prior, which is shown to be effective. When we build the fairness constraint based on the rating prediction performance, our framework cannot perform well on the fairness metric for the Top-N recommendation task. The above results demonstrate that all the components are necessary, and they play different roles to contribute to the final performance, where neither one of them can fully dominate the other one.

5.4 Influence of the Bias Level

Since our framework is designed to correct the dataset bias, an important problem is how different bias level influences the performance. To study this problem, we evaluate our framework under

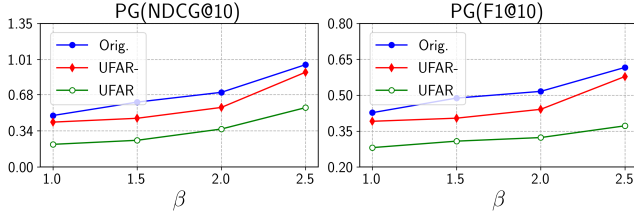


Figure 3: Influence of the bias level β . The results are percentage value with “%” omitted.

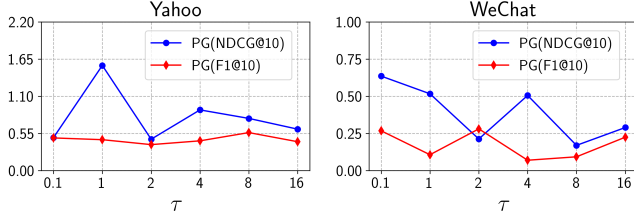


Figure 4: Influence of the temperature parameter τ . The results are percentage value with “%” omitted.

different biases. For flexibly tuning the dataset bias, we base the experiments on the WeChat dataset. Recall that in our dataset building process, the observation parameter β controls the bias level. To produce the datasets with different biases, we tune β in the range of $[1.0, 1.5, 2.0, 2.5]$, where a larger β means the dataset is more biased. The other settings follow the above experiment.

From the results shown in Figure 3, we can see: UFAR- performs better than the original model because it explicitly regularizes the performance gap between different user groups. UFAR can always achieve the best performance, which is consistent with the above experiments. By observing the changing trend of the performance, we find that all the models become worse as we increase the dataset bias. The reason can be that when the dataset is too biased, the observed information for the underrepresented items in the training set is too limited. All the models cannot learn effective patterns to predict them. As we increase the bias level, our framework can lead to larger performance improvement on all the metrics. This result is not surprising, since we leverage IPS to correct the fairness bias, which makes the underrepresented items significant to influence the group performance. By such bias correcting strategy, the training distribution in our framework is actually more consistent with the testing one compared with the other baselines, which helps to improve the performance.

5.5 Fairness on the Underrepresented Items

As claimed in the introduction, our framework should better reduce the performance gap between different user groups on the underrepresented items. To justify this claim, in this section, we evaluate our framework only based on the underrepresented items. In specific, for each user group, we firstly rank the items according to their interaction frequencies in the training set, where a higher rank means the item is more frequently interacted. Then we use the last 30% items in each user group for performance evaluation, while

Table 3: Fairness performance on underrepresented items.

	Yahoo				WeChat			
	Orig.	FOCF	UFAR-	UFAR	Orig.	FOCF	UFAR-	UFAR
PG(NDCG@10)	1.071	1.970	1.017	0.054	0.340	0.603	0.943	0.081
PG(F1@10)	0.099	0.358	0.405	0.071	0.078	0.455	0.260	0.004

the training set remains the same as the previous experiments. We follow the other experiment settings in section 5.3.

The results are presented in Table 3, where, for reference, we have also presented the performances of the original model and fairness aware baselines. We can see: compared with the original model, FOCF and UFAR-, our framework UFAR can significantly improve the fairness performance, that is, the performance gap on the underrepresented items can be greatly reduced. This observation is consistent on all the datasets and evaluation metrics, which suggests that our framework can indeed achieve the claims in the introduction. We speculate that the evaluated items only have a few number of interactions in the training set, thus if we treat each interaction equally, then the performance on these items can be sub-optimal due to the insufficient training. In our framework, we enhance the sample weights of these items to improve their importances, and for better reweighting, we design a range-based IPS strategy, which is shown to be effective. While both FOCF and UFAR- models have fairness constraints, their performances are actually not ideal without the bias correction. Moreover, this result has been observed in the previous experiment (see Table 2), but the performance gap is further amplified on the underrepresented items. All the results suggest that it is important to debias the fairness constraint for the observational datasets.

5.6 Parameter Analysis

In this section, we study the influence of hyper-parameters on the final fairness performance. More specifically, we focus on the temperature τ in the soft ranking metrics and the balancing weights λ and γ . The other experiment settings follow the above experiments.

Influence of τ . In our soft ranking mechanism, τ determines the softness of the probability that an item is ranked at a position. Extremely, if $\tau \rightarrow \infty$, then an item is uniformly ranked at different positions. When $\tau \rightarrow 0$, the soft-ranking mechanism is reduced to the traditional hard ranking metrics. In our experiment, we tune τ in the range of $[0.1, 1, 2, 4, 8, 16]$. From the results presented in Figure 4, we can see: the performance fluctuates severely as we use different τ 's, and the changing patterns highly depend on the datasets and evaluation metrics. A consistent observation is that the best performance is usually achieved when τ is moderate. Too large and small τ 's are both sub-optimal. We speculate that too large τ may lower the ranking significance between different positions, which may make the model hard to separate the positive and negative samples. However, if we set τ as a too small value, the model may rely on the observed dataset too much, which can also be inferior, since the observed dataset may contain too much noise and randomness. The above results suggest that τ is a sensitive parameter for our framework, and in practice, it is important to tune it carefully for achieving satisfying performance.

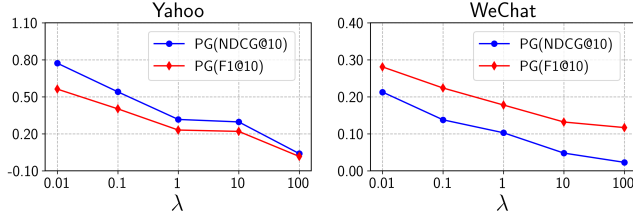


Figure 5: Influence of the balancing parameter λ . The results are percentage value with “%” omitted.

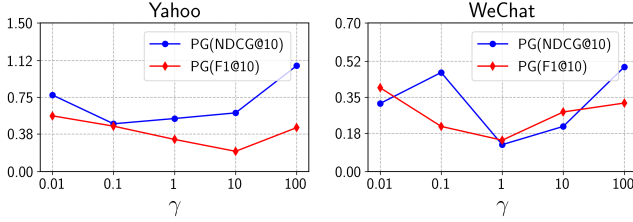


Figure 6: Influence of the balancing parameter γ . The results are percentage value with “%” omitted.

Influence of λ and γ . In our objective, λ and γ determine how to balance the recommendation objective with the fairness constraint and IPS regularization term. We tune λ and γ in the range of $[0.01, 0.1, 1, 10, 100]$. The results are presented in Figure 5 and 6, where we can see: as we increase λ , the performance becomes better on all the datasets and evaluation metrics. This is as expected, since λ controls the importance of the fairness constraint. If we put more focus on the fairness optimization, then the performance gap can be lowered. For γ , the best performance is usually achieved when it is a moderate value. We speculate that if γ is too large, then the IPS may have very small exploration spaces, and its value can be very similar to the initially estimated IPS, which can be inaccurate due to the noisy recommendation dataset. If γ is set as a too small value, then the regularization on the IPS can be not sufficient, and the initially estimated value may have little effect, which makes it hard to efficiently discover the true IPS.

6 CONCLUSION AND FUTURE WORK

In this paper, we propose an unbiased fairness aware Top-N recommendation framework. The unbiasedness is achieved based on IPS, and to improve its accuracy, we introduce a range-based strategy. For building the fairness constraint, we soften the ranking metrics to evaluate the performance gap between different user groups. Extensive experiments are conducted to verify the effectiveness of our framework. This paper actually makes a first step towards unbiased fairness aware recommendation. There is still much room left for improvement. For example, one can extend our framework to multiple sensitive features or multi-value sensitive attributes. In addition, it is also interesting to extend our framework to the sequential and graph settings, where the current item is also biased by the neighbor users or items.

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REFERENCES

- [1] Quanyu Dai, Zhenhua Dong, and Xu Chen. 2022. Debaised recommendation with neural stratification. *AI Open* 3 (2022), 213–217.
- [2] Quanyu Dai, Haoxuan Li, Peng Wu, Zhenhua Dong, Xiao-Hua Zhou, Rui Zhang, Rui Zhang, and Jie Sun. 2022. A generalized doubly robust learning framework for debiasing post-click conversion rate prediction. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*.
- [3] Wei Du and Xintao Wu. 2021. Robust fairness-aware learning under sample selection bias. *Proceedings of 7th ACM International Conference on Learning Representations* (2021).
- [4] Yingqiang Ge, Shuchang Liu, Ruoyuan Gao, Yikun Xian, Yunqi Li, Xiangyu Zhao, Changhua Pei, Fei Sun, Junfeng Ge, Wenwu Ou, et al. 2021. Towards long-term fairness in recommendation. In *Proceedings of the 14th ACM International Conference on Web Search and Data Mining*.
- [5] Yingqiang Ge, Juntao Tan, Yan Zhu, Yinglong Xia, Jiebo Luo, Shuchang Liu, Zuohui Fu, Shijie Geng, Zelong Li, and Yongfeng Zhang. 2022. Explainable fairness in recommendation. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 681–691.
- [6] Aditya Grover, Eric Wang, Aaron Zweig, and Stefano Ermon. 2019. Stochastic optimization of sorting networks via continuous relaxations. *Proceedings of the 7th ACM International Conference on Learning Representations* (2019).
- [7] Siyuan Guo, Lixin Zou, Yiding Liu, Wenwen Ye, Suqi Cheng, Shuaiqiang Wang, Hechang Chen, Dawei Yin, and Yi Chang. 2021. Enhanced Doubly Robust Learning for Debiasing Post-Click Conversion Rate Estimation. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*.
- [8] Xiangnan He, Kuan Deng, Xiang Wang, Yan Li, Yongdong Zhang, and Meng Wang. 2020. Lightgcn: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval*.
- [9] Xiangnan He, Lizi Liao, Hanwang Zhang, Liqiang Nie, Xia Hu, and Tat-Seng Chua. 2017. Neural collaborative filtering. In *Proceedings of the 26th international conference on world wide web*.
- [10] Rashidul Islam, Kamrun Naher Keya, Ziqian Zeng, Shimei Pan, and James Foulds. 2021. Debiasing Career Recommendations with Neural Fair Collaborative Filtering. In *Proceedings of the Web Conference 2021*.
- [11] Christopher C Johnson. 2014. Logistic matrix factorization for implicit feedback data. *Advances in Neural Information Processing Systems* (2014).
- [12] Toshihiro Kamishima, Shotaro Akaho, Hideki Asoh, and Jun Sakuma. 2018. Recommendation independence. In *Conference on fairness, accountability and transparency*.
- [13] Diederik P. Kingma and Jimmy Ba. 2014. *Proceedings of the 3rd International Conference on Learning Representations*.
- [14] Yehuda Koren, Robert Bell, and Chris Volinsky. 2009. Matrix factorization techniques for recommender systems. *Computer* (2009).
- [15] Hyunsung Lee, Sangwoo Cho, Yeongjae Jang, Jaekwang Kim, and Honguk Woo. 2021. Differentiable Ranking Metric Using Relaxed Sorting for Top-K Recommendation. *IEEE Access* (2021).
- [16] Jie Li, Yongli Ren, and Ke Deng. 2022. FairGAN: GANs-Based Fairness-Aware Learning for Recommendations with Implicit Feedback. In *Proceedings of the ACM Web Conference 2022*.
- [17] Roger Zhe Li, Julián Urbano, and Alan Hanjalic. 2021. Leave no user behind: Towards improving the utility of recommender systems for non-mainstream users. In *Proceedings of the 14th ACM International Conference on Web Search and Data Mining*.
- [18] Yunqi Li, Hanxiong Chen, Zuohui Fu, Yingqiang Ge, and Yongfeng Zhang. 2021. User-oriented fairness in recommendation. In *Proceedings of the Web Conference 2021*.

- [19] Yunqi Li, Hanxiong Chen, Shuyuan Xu, Yingqiang Ge, and Yongfeng Zhang. 2021. Towards personalized fairness based on causal notion. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*.
- [20] Yunqi Li, Hanxiong Chen, Shuyuan Xu, Yingqiang Ge, and Yongfeng Zhang. 2021. Towards Personalized Fairness Based on Causal Notion.
- [21] Mohammadmehdi Naghiaei, Hossein A Rahmani, and Yashar Deldjoo. 2022. Cpfair: Personalized consumer and producer fairness re-ranking for recommender systems. *arXiv preprint arXiv:2204.08085* (2022).
- [22] Yuta Saito. 2020. Asymmetric tri-training for debiasing missing-not-at-random explicit feedback. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*.
- [23] Yuta Saito, Gota Morishta, and Shota Yasui. 2020. Dual Learning Algorithm for Delayed Conversions.
- [24] Yuta Saito, Suguru Yaginuma, Yuta Nishino, Hayato Sakata, and Kazuhide Nakata. 2020. Unbiased recommender learning from missing-not-at-random implicit feedback. In *Proceedings of the 13th ACM International Conference on Web Search and Data Mining*.
- [25] Tobias Schnabel, Adith Swaminathan, Ashudeep Singh, Navin Chandak, and Thorsten Joachims. 2016. Recommendations as treatments: Debiasing learning and evaluation. In *international conference on machine learning*.
- [26] Xiaojie Wang, Rui Zhang, Yu Sun, and Jianzhong Qi. 2019. Doubly Robust Joint Learning for Recommendation on Data Missing Not at Random. In *Proceedings of the 36th International Conference on Machine Learning*.
- [27] Yifan Wang, Weizhi Ma, Min Zhang*, Yiqun Liu, and Shaoping Ma. 2022. A Survey on the Fairness of Recommender Systems. *ACM Trans. Inf. Syst.* (2022).
- [28] Zhenlei Wang, Shiqi Shen, Zhipeng Wang, Bo Chen, Xu Chen, and Ji-Rong Wen. 2022. Unbiased sequential recommendation with latent confounders. In *Proceedings of the ACM Web Conference 2022*. 2195–2204.
- [29] Zhenlei Wang, Jingsen Zhang, Hongteng Xu, Xu Chen, Yongfeng Zhang, Wayne Xin Zhao, and Ji-Rong Wen. 2021. Counterfactual data-augmented sequential recommendation. In *Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval*. 347–356.
- [30] Tianxin Wei and Jingrui He. 2022. Comprehensive Fair Meta-Learned Recommender System. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*.
- [31] Chuhan Wu, Fangzhao Wu, Xiting Wang, Yongfeng Huang, and Xing Xie. 2021. Fairness-aware news recommendation with decomposed adversarial learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- [32] Le Wu, Lei Chen, Pengyang Shao, Richang Hong, Xiting Wang, and Meng Wang. 2021. Learning fair representations for recommendation: A graph-based perspective. In *Proceedings of the Web Conference 2021*. 2198–2208.
- [33] Yiqing Wu, Ruobing Xie, Yongchun Zhu, Fuzhen Zhuang, Ao Xiang, Xu Zhang, Leyu Lin, and Qing He. 2022. Selective fairness in recommendation via prompts. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2657–2662.
- [34] Chen Xu, Jun Xu, Xu Chen, Zhenghua Dong, and Ji-Rong Wen. 2022. Dually Enhanced Propensity Score Estimation in Sequential Recommendation. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*. 2260–2269.
- [35] Longqi Yang, Yin Cui, Yuan Xuan, Chenyang Wang, Serge Belongie, and Deborah Estrin. 2018. Unbiased offline recommender evaluation for missing-not-at-random implicit feedback. In *Proceedings of the 12th ACM conference on recommender systems*.
- [36] Mengyue Yang, Guohao Cai, Furui Liu, Jiarui Jin, Zhenhua Dong, Xiuqiang He, Jianye Hao, Weiqi Shao, Jun Wang, and Xu Chen. 2023. Debaised recommendation with user feature balancing. *ACM Transactions on Information Systems* 41, 4 (2023), 1–25.
- [37] Sirui Yao and Bert Huang. 2017. Beyond parity: Fairness objectives for collaborative filtering. *Advances in neural information processing systems* (2017).
- [38] Jingsen Zhang, Xu Chen, Jiakai Tang, Weiqi Shao, Quanyu Dai, Zhenhua Dong, and Rui Zhang. 2023. Recommendation with Causality enhanced Natural Language Explanations. In *Proceedings of the ACM Web Conference 2023*. 876–886.
- [39] Xun Zheng, Bryon Aragam, Pradeep K Ravikumar, and Eric P Xing. 2018. DAGs with NO TEARS: Continuous Optimization for Structure Learning. In *Advances in Neural Information Processing Systems*.
- [40] Ziwei Zhu, Xia Hu, and James Caverlee. 2018. Fairness-Aware Tensor-Based Recommendation. In *Proceedings of the 27th ACM International Conference on Information and Knowledge Management*.
- [41] Ziwei Zhu, Jianling Wang, and James Caverlee. 2020. Measuring and mitigating item under-recommendation bias in personalized ranking systems. In *Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval*. 449–458.